

Chapter 10: Wage Structures Across Markets

ECON 300 - Labour Economics

Muhebullah Karimzada

Winter 2026

- Understand the Roy model with heterogeneous skills and market sorting.
- Explain wage differences across labour markets via supply, demand, and institutions.
- Compare wages across occupations, industries, regions, firm size, and sectors.
- Interpret interindustry differentials through the lens of efficiency wages.
- Discuss why the public sector often pays more than the private sector.

- Understand the Roy model with heterogeneous skills and market sorting.
- Explain wage differences across labour markets via supply, demand, and institutions.
- Compare wages across occupations, industries, regions, firm size, and sectors.
- Interpret interindustry differentials through the lens of efficiency wages.
- Discuss why the public sector often pays more than the private sector.

- Understand the Roy model with heterogeneous skills and market sorting.
- Explain wage differences across labour markets via supply, demand, and institutions.
- Compare wages across occupations, industries, regions, firm size, and sectors.
- Interpret interindustry differentials through the lens of efficiency wages.
- Discuss why the public sector often pays more than the private sector.

- Understand the Roy model with heterogeneous skills and market sorting.
- Explain wage differences across labour markets via supply, demand, and institutions.
- Compare wages across occupations, industries, regions, firm size, and sectors.
- Interpret interindustry differentials through the lens of efficiency wages.
- Discuss why the public sector often pays more than the private sector.

- Understand the Roy model with heterogeneous skills and market sorting.
- Explain wage differences across labour markets via supply, demand, and institutions.
- Compare wages across occupations, industries, regions, firm size, and sectors.
- Interpret interindustry differentials through the lens of efficiency wages.
- Discuss why the public sector often pays more than the private sector.

- Why do wages vary across labour markets even for apparently identical workers?
- Are public sector workers overpaid?
- Why are wages higher in Alberta than in Ontario?
- What drives interprovincial migration decisions?
- Do higher wages at large firms contradict the neoclassical model?
- Which industries/occupations pay the most—and is it due to worker skills?

- Why do wages vary across labour markets even for apparently identical workers?
- Are public sector workers overpaid?
- Why are wages higher in Alberta than in Ontario?
- What drives interprovincial migration decisions?
- Do higher wages at large firms contradict the neoclassical model?
- Which industries/occupations pay the most—and is it due to worker skills?

- Why do wages vary across labour markets even for apparently identical workers?
- Are public sector workers overpaid?
- Why are wages higher in Alberta than in Ontario?
- What drives interprovincial migration decisions?
- Do higher wages at large firms contradict the neoclassical model?
- Which industries/occupations pay the most—and is it due to worker skills?

- Why do wages vary across labour markets even for apparently identical workers?
- Are public sector workers overpaid?
- Why are wages higher in Alberta than in Ontario?
- What drives interprovincial migration decisions?
- Do higher wages at large firms contradict the neoclassical model?
- Which industries/occupations pay the most—and is it due to worker skills?

- Why do wages vary across labour markets even for apparently identical workers?
- Are public sector workers overpaid?
- Why are wages higher in Alberta than in Ontario?
- What drives interprovincial migration decisions?
- Do higher wages at large firms contradict the neoclassical model?
- Which industries/occupations pay the most—and is it due to worker skills?

- Why do wages vary across labour markets even for apparently identical workers?
- Are public sector workers overpaid?
- Why are wages higher in Alberta than in Ontario?
- What drives interprovincial migration decisions?
- Do higher wages at large firms contradict the neoclassical model?
- Which industries/occupations pay the most—and is it due to worker skills?

Individual (log) wages depend on schooling, experience, ability/luck, and other traits:

$$\ln Y_i = \alpha + rS_i + \beta_1 \text{EXP}_i + \beta_2 \text{EXP}_i^2 + \gamma X_i + \varepsilon_i$$

- Y : earnings; S : years of schooling; r : return to schooling (IRR interpretation);
- EXP : experience proxy; X : controls; ε : unobservables.

Earnings Model: Returns to Traits

Individual (log) wages depend on schooling, experience, ability/luck, and other traits:

$$\ln Y_i = \alpha + rS_i + \beta_1 \text{EXP}_i + \beta_2 \text{EXP}_i^2 + \gamma X_i + \varepsilon_i$$

- Y : earnings; S : years of schooling; r : return to schooling (IRR interpretation);
- EXP : experience proxy; X : controls; ε : unobservables.

Pure Regional, Occupational, and Industry Premia

- Conditional wage differences remain after adjusting for S , EXP , X .
- “Premiums” summarize province, occupation, and industry effects.

- Conditional wage differences remain after adjusting for S , EXP , X .
- “Premiums” summarize province, occupation, and industry effects.

Table 10.1 — Earnings Differentials by Province, Canada, 2015

Province	% Jobs	Premium
Newfoundland and Labrador	1.3	4.1
Prince Edward Island	0.3	−12.8
Nova Scotia	2.7	−8.1
New Brunswick	2.1	−11.1
Quebec	22.8	−11.5
Ontario	39.9	—
Manitoba	3.5	−2.6
Saskatchewan	2.8	10.1
Alberta	13.1	18.1
British Columbia	12.1	1.6

Table 10.1 — Earnings Differentials by Occupation and Industry, Canada, 2015

Occupation			Industry		
Group	% Jobs	Premium	Group	% Jobs	Premium
Senior management	2.0	56.1	Agriculture, forestry, and fishing	1.5	−35.1
Middle management	5.0	28.5	Other primary	2.4	51.3
Other management	9.1	9.5	Utilities	1.5	38.5
Professionals (business/finance)	3.4	14.8	Construction	10.2	−4.2
Finance and administration	5.8	−15.6	Manufacturing	16.2	—
Clerical	1.1	−25.6	Wholesale trade	6.4	1.7
Professionals (science)	8.0	14.2	Retail trade	9.6	−23.6
Other science occupations	5.8	−2.1	Transportation/warehousing	7.1	−0.7
Professionals (health)	1.0	30.9	Information and culture	2.9	5.4
Other health occupations	1.0	0.9	Finance and insurance	4.5	16.2
Professionals (education/social)	4.8	10.4	Real estate and leasing	1.3	−8.9
Other education/social	3.0	10.0	Professional services	7.9	−3.1
Professionals (arts/culture/rec)	1.4	−21.3	Admin/support, waste mgmt	3.5	−25.7
Sales and service supervisors	4.4	−9.5	Educational services	4.5	−16.4
Sales representatives	5.7	−18.7	Health and social assistance	3.5	−22.8
Sales and services support	4.2	−33.4	Accommodation and food	3.6	−57.6
Trade workers	15.3	—	Other services (excl. public admin)	4.6	−27.8
Transportation/trade helpers	10.1	−22.7	Public administration	8.8	6.8

- National Occupational Classification (NOC): 26 two-digit major groups; 17 broad classes; hundreds of detailed codes.
- Occupational wage gaps reflect nonpecuniary traits, human capital, endowed skills, demand shifts, and institutions.

- National Occupational Classification (NOC): 26 two-digit major groups; 17 broad classes; hundreds of detailed codes.
- Occupational wage gaps reflect nonpecuniary traits, human capital, endowed skills, demand shifts, and institutions.

Figure 10.1 — Occupational Wage Differentials (*Placeholder*)

Figure 10.1 — Comparative wage levels by occupation.
Insert figure/diagram here later.

Why Do Occupations Pay Differently?

- **Compensating differentials:** risk, conditions, schedule.
- **Human capital:** training requirements and entry costs.
- **Endowments:** innate abilities valued differently across jobs.
- **Short-run demand:** cyclical/sectoral shifts.
- **Noncompetitive factors:** licensing, entry limits, unions, wage laws.

Why Do Occupations Pay Differently?

- **Compensating differentials:** risk, conditions, schedule.
- **Human capital:** training requirements and entry costs.
- **Endowments:** innate abilities valued differently across jobs.
- **Short-run demand:** cyclical/sectoral shifts.
- **Noncompetitive factors:** licensing, entry limits, unions, wage laws.

Why Do Occupations Pay Differently?

- **Compensating differentials:** risk, conditions, schedule.
- **Human capital:** training requirements and entry costs.
- **Endowments:** innate abilities valued differently across jobs.
- **Short-run demand:** cyclical/sectoral shifts.
- **Noncompetitive factors:** licensing, entry limits, unions, wage laws.

Why Do Occupations Pay Differently?

- **Compensating differentials:** risk, conditions, schedule.
- **Human capital:** training requirements and entry costs.
- **Endowments:** innate abilities valued differently across jobs.
- **Short-run demand:** cyclical/sectoral shifts.
- **Noncompetitive factors:** licensing, entry limits, unions, wage laws.

Why Do Occupations Pay Differently?

- **Compensating differentials:** risk, conditions, schedule.
- **Human capital:** training requirements and entry costs.
- **Endowments:** innate abilities valued differently across jobs.
- **Short-run demand:** cyclical/sectoral shifts.
- **Noncompetitive factors:** licensing, entry limits, unions, wage laws.

The Roy Model: Sorting by Comparative Advantage

- Workers have different skill vectors; wages differ by match quality across occupations.
- Individuals self-select into the occupation yielding the highest wage for their skill bundle.
- Implication: observed wage distributions reflect **sorting**, not just treatment effects.

The Roy Model: Sorting by Comparative Advantage

- Workers have different skill vectors; wages differ by match quality across occupations.
- Individuals self-select into the occupation yielding the highest wage for their skill bundle.
- Implication: observed wage distributions reflect **sorting**, not just treatment effects.

The Roy Model: Sorting by Comparative Advantage

- Workers have different skill vectors; wages differ by match quality across occupations.
- Individuals self-select into the occupation yielding the highest wage for their skill bundle.
- Implication: observed wage distributions reflect **sorting**, not just treatment effects.

- Reasons: geographic preferences; compensating differences (cost of living, remoteness, climate, congestion, pollution).
- Short run: disequilibrium encourages mobility; long run: convergence via flows of labour/capital.
- Frictions: moving costs, policy barriers (e.g., licensing), and networks.

- Reasons: geographic preferences; compensating differences (cost of living, remoteness, climate, congestion, pollution).
- Short run: disequilibrium encourages mobility; long run: convergence via flows of labour/capital.
- Frictions: moving costs, policy barriers (e.g., licensing), and networks.

- Reasons: geographic preferences; compensating differences (cost of living, remoteness, climate, congestion, pollution).
- Short run: disequilibrium encourages mobility; long run: convergence via flows of labour/capital.
- Frictions: moving costs, policy barriers (e.g., licensing), and networks.

Migration Decision: Benefits vs. Costs

- Move if present value of wage/personal benefits exceeds moving and adaptation costs.
- Influences: age, unemployment rates, business cycle, distance, cultural differences.

Migration Decision: Benefits vs. Costs

- Move if present value of wage/personal benefits exceeds moving and adaptation costs.
- Influences: age, unemployment rates, business cycle, distance, cultural differences.

Oil shock — Alberta wage premium; spillovers to construction/services; migration inflows.

Insert figure/diagram here later.

What Drives Interindustry Wage Gaps?

- Composition: occupation mix and worker traits.
- Region dominance and product-market forces (trade, technology, deregulation).
- Nonpecuniary aspects: conditions, seasonality, cyclicalities.
- Noncompetitive factors: monopoly rents, unions, wage floors, licensing.

What Drives Interindustry Wage Gaps?

- Composition: occupation mix and worker traits.
- Region dominance and product-market forces (trade, technology, deregulation).
- Nonpecuniary aspects: conditions, seasonality, cyclicalities.
- Noncompetitive factors: monopoly rents, unions, wage floors, licensing.

What Drives Interindustry Wage Gaps?

- Composition: occupation mix and worker traits.
- Region dominance and product-market forces (trade, technology, deregulation).
- Nonpecuniary aspects: conditions, seasonality, cyclicalities.
- Noncompetitive factors: monopoly rents, unions, wage floors, licensing.

What Drives Interindustry Wage Gaps?

- Composition: occupation mix and worker traits.
- Region dominance and product-market forces (trade, technology, deregulation).
- Nonpecuniary aspects: conditions, seasonality, cyclicalities.
- Noncompetitive factors: monopoly rents, unions, wage floors, licensing.

Efficiency Wages and “Good Jobs”

- Firms may pay above-market wages to improve morale, deter shirking, reduce turnover, and discourage unionization.
- Higher wages can create applicant queues and persistent interindustry premia.
- Competitive pressure (imports, deregulation, more rivals) tends to compress premia.

Efficiency Wages and “Good Jobs”

- Firms may pay above-market wages to improve morale, deter shirking, reduce turnover, and discourage unionization.
- Higher wages can create applicant queues and persistent interindustry premia.
- Competitive pressure (imports, deregulation, more rivals) tends to compress premia.

- Firms may pay above-market wages to improve morale, deter shirking, reduce turnover, and discourage unionization.
- Higher wages can create applicant queues and persistent interindustry premia.
- Competitive pressure (imports, deregulation, more rivals) tends to compress premia.

- Interindustry wage structure is **stable** over time and across countries.
- Product-market competition lowers wages relative to protected industries.
- Pure interindustry differentials (rents) persist—consistent with efficiency wages.

- Interindustry wage structure is **stable** over time and across countries.
- Product-market competition lowers wages relative to protected industries.
- Pure interindustry differentials (rents) persist—consistent with efficiency wages.

- Interindustry wage structure is **stable** over time and across countries.
- Product-market competition lowers wages relative to protected industries.
- Pure interindustry differentials (rents) persist—consistent with efficiency wages.

- Larger firms often pay more—due to rents, internal labor markets, scale economies, and sorting.
- Also reflect nonpecuniary factors and union presence in larger firms.

- Larger firms often pay more—due to rents, internal labor markets, scale economies, and sorting.
- Also reflect nonpecuniary factors and union presence in larger firms.

- Public sector: about one-fifth of Canadian employment; higher wages and benefits on average.
- Reasons: job security, benefit generosity, political constraints, monopsony power, unionization, and demand inelasticity.

- Public sector: about one-fifth of Canadian employment; higher wages and benefits on average.
- Reasons: job security, benefit generosity, political constraints, monopsony power, unionization, and demand inelasticity.

Figure 10.2 — Public sector employment share over time.
Insert figure/diagram here later.

- Wage structure has multiple dimensions: occupation, industry, region, firm size, and sector.
- Roy model explains sorting and the mapping from heterogeneous skills to observed wages.
- Interindustry and interfirm differentials persist due to rents, institutions, and compensating differences.
- Public sector typically pays more; interpretation requires adjusting for traits and benefits.

- Wage structure has multiple dimensions: occupation, industry, region, firm size, and sector.
- Roy model explains sorting and the mapping from heterogeneous skills to observed wages.
- Interindustry and interfirm differentials persist due to rents, institutions, and compensating differences.
- Public sector typically pays more; interpretation requires adjusting for traits and benefits.

- Wage structure has multiple dimensions: occupation, industry, region, firm size, and sector.
- Roy model explains sorting and the mapping from heterogeneous skills to observed wages.
- Interindustry and interfirm differentials persist due to rents, institutions, and compensating differences.
- Public sector typically pays more; interpretation requires adjusting for traits and benefits.

- Wage structure has multiple dimensions: occupation, industry, region, firm size, and sector.
- Roy model explains sorting and the mapping from heterogeneous skills to observed wages.
- Interindustry and interfirm differentials persist due to rents, institutions, and compensating differences.
- Public sector typically pays more; interpretation requires adjusting for traits and benefits.

Thank You!